Mock Interview Edition

People You May Know: Graph Candidates, Two-Pass Ranking and Equity

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

Jump to the contents | Round plans | Question sheets | Score sheet

HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

Contents

Every line below is a link: click it to jump to that page. The outline panel of your PDF viewer lists the same entries, and the “Contents” link in every page footer brings you back here. Question numbers are the same as in the Study Edition.

Sections

Section A Round plans: 5 ready-made sessions 3

Section B Question sheets: every scenario with probes and notes boxes 4

Section C Score sheet 30

Parts at a glance (question sheets)

1 Framing the problem: the objective, the labels and the size of the graph 4

2 Data and candidates: edge types, triangle closing and cold start 9

3 Ranking and equity: two passes, calibration, impression rules and fairness 15

4 Serving, evaluation and operations 22

Every scenario, by part

Part 1 Framing the problem: the objective, the labels and the size of the graph 4

01 What does PYMK predict? Framing 4

02 Size the edge table Estimation 5

03 Second-degree expansion Estimation 7

04 How many nodes does the daily batch need? Throughput 8

Part 2 Data and candidates: edge types, triangle closing and cold start 9

05 Why not one ranking model for every edge? Data 9

06 What does triangle closing find and miss? Candidates 11

07 A member with no connections Cold start 12

08 Review the friends-of-friends function Code review 13

Part 3 Ranking and equity: two passes, calibration, impression rules and fairness 15

09 Two passes, and why they are independent Architecture 15

10 Why calibrate, and what does it not fix? Calibration 16

11 Importance factors and impression guarantees Ranking 18

12 Discount the overloaded recipient Re-ranking 19

13 Read the fairness result Evaluation 20

Part 4 Serving, evaluation and operations 22

14 Why recompute everything every day? Serving 22

15 Cost one member's expansion Scale 23

16 Review the connection table Code review 25

17 Measure the recipient side Evaluation 26

18 Risks, and ten times the graph Risk 28

SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

Round 1 Objective and sizing 4

01 What does PYMK predict? Framing 4
02 Size the edge table Estimation 5
03 Second-degree expansion Estimation 7
04 How many nodes does the daily batch need? Throughput 8
17 Measure the recipient side Evaluation 26

Round 2 Data and candidates 9

05 Why not one ranking model for every edge? Data 9
06 What does triangle closing find and miss? Candidates 11
07 A member with no connections Cold start 12
08 Review the friends-of-friends function Code review 13
03 Second-degree expansion Estimation 7

Round 3 Ranking and equity 15

09 Two passes, and why they are independent Architecture 15
10 Why calibrate, and what does it not fix? Calibration 16
11 Importance factors and impression guarantees Ranking 18
12 Discount the overloaded recipient Re-ranking 19
13 Read the fairness result Evaluation 20

Round 4 Serving, evaluation and operations 22

14 Why recompute everything every day? Serving 22
15 Cost one member's expansion Scale 23
16 Review the connection table Code review 25
17 Measure the recipient side Evaluation 26
18 Risks, and ten times the graph Risk 28

Round 5 System design loop (mixed) 4

01 What does PYMK predict? Framing 4
06 What does triangle closing find and miss? Candidates 11
09 Two passes, and why they are independent Architecture 15
13 Read the fairness result Evaluation 20
15 Cost one member's expansion Scale 23

SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: the objective, the labels and the size of the graph

Saying exactly what is predicted, separating events that sound alike, and sizing a graph and its second-degree expansion from stated assumptions.

01 What does PYMK predict?

FRAMING

SCENARIO

A colleague summarizes the LinkedIn pages: PYMK predicts connections, the second-pass ranker predicts connections too, and labels are connected pairs in the graph. Check each statement against the sources.

OPENING PROBE

What does the team page say?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the second-pass ranker predict?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the pages say about labels?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How does a connection form?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 Size the edge table

ESTIMATION

SCENARIO

You must store the connection graph as an edge table with one row per directed half, so that a member's connections come from one shard. Assume 700 million members and an average of 400 connections, rows of 24 bytes and 2,048 shards. A reviewer checks your arithmetic.

OPENING PROBE

How many rows?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How big is it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Does capacity force 2,048 shards?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would change with a different degree?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Second-degree expansion

ESTIMATION

SCENARIO

The team page says PYMK processes hundreds of billions of potential connections a day and that, because it looks at the second-degree network, processing grows much faster than the site. With 700 million members and 400 connections each, check whether that is consistent.

OPENING PROBE

How many second-degree walks are there?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How does that compare with the page?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What are the readings?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does growth faster than the site mean?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 How many nodes does the daily batch need?

THROUGHPUT

SCENARIO

A daily job expands friends of friends by reading each friend's connection list for each member. Using the Question 2 sizing, find the load, and say what sets the number of shards.

OPENING PROBE

How many lookups?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How much data comes back?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How many nodes?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

So what are 2,048 shards for?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Data and candidates: edge types, triangle closing and cold start

Why different edges need different signals, what triangle closing finds and misses, and what a graph feature cannot do for a member with no edges.

05 Why not one ranking model for every edge?

DATA

SCENARIO

A colleague proposes one model to recommend connections, hashtags to follow, groups and events, trained on all the interaction data together. The 2020 post separates them. Explain why, using the post's own points.

OPENING PROBE

How do the edge types differ?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the training data over-represent?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the post do about it?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What about the models themselves?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 What does triangle closing find and miss?

CANDIDATES

SCENARIO

A teammate says: triangle closing gives us candidates that are precise and easy to explain, so we should use only that. The team page says it then scores the closed triangles with other features. Say why.

OPENING PROBE

What is the signal?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does it miss?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What do the other features do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Which candidate sources would you add?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 A member with no connections

COLD START

SCENARIO

A new member joins with no connections. Triangle closing finds nothing to expand. The 2020 post notes that inactive members lead to lack of data and lower-quality recommendations. Design the first week.

OPENING PROBE

Why is there nothing to expand?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What signals exist on day one?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

How would candidates be proposed?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What should you measure?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the friends-of-friends function

CODE REVIEW

SCENARIO

A colleague writes the function below. In a test it returns the member's own id, people already connected to them, and a huge dictionary, and one run on a member who is a friend of a very popular account never finishes. Find the faults.

```
def friends_of_friends(member, adj):
    mutual = {}
    for friend in adj[member]:
        for other in adj[friend]:
            mutual[other] = mutual.get(other, 0) + 1
    return mutual
```

OPENING PROBE

What does it wrongly return?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What about a member not in the graph?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What about the popular account?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What about the output?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

Ranking and equity: two passes, calibration, impression rules and fairness

How scores from different models are made comparable, how guarantees and importance differ, and how to read results about a re-ranker and a fairness change.

09 Two passes, and why they are independent

ARCHITECTURE

SCENARIO

Seven teams each own a first-pass model for one entity type, and one team owns the second-pass ranker. A manager asks why the second pass is not trained end to end with the first-pass models, which might be a little more accurate.

OPENING PROBE

What do the passes do?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What would coupling do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the post choose?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How is the second pass trained?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

10 Why calibrate, and what does it not fix?

CALIBRATION

SCENARIO

The connection model gives a pair the score 0.40, and the hashtag-follow model gives a hashtag the score 0.40. The second pass must compare them. The post calibrates each first-pass score. Show what that does and what it leaves undone.

OPENING PROBE

What is the problem with raw scores?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does calibration do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is still different after calibration?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why does that target help?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

11 Importance factors and impression guarantees

RANKING

SCENARIO

A review note says: the post's 1:2 example means connections are weighted twice as much as follows. Check the note against the post.

OPENING PROBE

What are the two mechanisms?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the 1:2 example say?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Compute it.

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Why have guarantees at all?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

12 Discount the overloaded recipient

RE-RANKING

SCENARIO

Influencers, executives and recruiters receive huge numbers of invitations. The 2021 post multiplies a candidate's score by a decay factor. Show how it changes a ranking, and say what the post reports and why it expected a loss.

OPENING PROBE

What is the rule?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Apply it.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did the team expect?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What happened?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

13 Read the fairness result

EVALUATION

SCENARIO

The 2021 post re-ranks connection suggestions for equality of opportunity and equalized odds. It reports a 5.44% rise in invitations sent to infrequent members, 4.8% more connections made by them, and neutral metrics for frequent members. A director says this proves nobody was harmed. Respond.

OPENING PROBE

What does each number count?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How does the post read the neutral FM metrics?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Why is that not proof of no harm?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What would you ask for?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

PART 4

Serving, evaluation and operations

Why recomputing daily can be the right call, what a sharded graph costs per member, and reading results and risks.

14 Why recompute everything every day?

SERVING

SCENARIO

The team page says new PYMK results are pushed every day. An engineer proposes updating candidates incrementally as each connection forms. Size the proposal.

OPENING PROBE

What does a new connection change?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How many changes arrive a day?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does one member see?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the daily batch cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 20

15 Cost one member's expansion

SCALE

SCENARIO

Edges are stored in a table sharded by the owner member's id, with each connection stored twice. Cost one member's friends-of-friends expansion, and say how to avoid it.

OPENING PROBE

How many queries?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does that cost for the batch?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the alternative?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does a hub do?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

16 Review the connection table

CODE REVIEW

SCENARIO

A colleague stores each connection once, ordered as (low id, high id), and shards by the low id to halve storage. Looking up a member's friends is slow and sometimes incomplete in a test. Find the faults.

```
CREATE TABLE connections (
    low_id  BIGINT NOT NULL,
    high_id BIGINT NOT NULL,
    PRIMARY KEY (low_id, high_id)          -- sharded by low_id
);

-- friends of X
SELECT high_id FROM connections WHERE low_id = :x
UNION
SELECT low_id  FROM connections WHERE high_id = :x;
```

OPENING PROBE

What does a friends lookup need?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Where do the rows live?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the table save?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

17 Measure the recipient side

EVALUATION

SCENARIO

The team says a change that sends fewer invitations improved overall sessions. A reviewer asks how they could attribute sessions of people who received invitations from several senders, and what the pages say about offline metrics.

OPENING PROBE

Why is the recipient side hard?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What is the rule?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What is the total effect of the decay change?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What do the pages say about offline metrics?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

How do you monitor groups?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

18 Risks, and ten times the graph

RISK

SCENARIO

A launch review asks what can go wrong with PYMK and what grows if the graph grows ten times.

OPENING PROBE

What risks do the pages name?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does ten times members do?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does ten times degree do?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does not scale with traffic?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	What does PYMK predict?	5			
02	Size the edge table	5			
03	Second-degree expansion	5			
04	How many nodes does the daily batch need?	5			
05	Why not one ranking model for every edge?	5			
06	What does triangle closing find and miss?	5			
07	A member with no connections	5			
08	Review the friends-of-friends function	6			
09	Two passes, and why they are independent	5			
10	Why calibrate, and what does it not fix?	5			
11	Importance factors and impression guarantees	5			
12	Discount the overloaded recipient	5			
13	Read the fairness result	5			
14	Why recompute everything every day?	5			
15	Cost one member's expansion	5			
16	Review the connection table	5			
17	Measure the recipient side	6			
18	Risks, and ten times the graph	5			